Psyche Mission Dictionary

NASA Planetary Data System

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USER GUIDE

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The Psyche Mission Dictionary (psyche) contains classes, attributes and rules specific to the Psyche mission, its instruments and working group results.

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CHAPTER

ONE

INTRODUCTION

1. Purpose of this User's Guide
2. Audience
3. Applicable Documents

CHAPTER

TWO

OVERVIEW OF THE PSYCHE MISSION LOCAL DATA DICTIONARY

The Psyche Local Data Dictionary (LDD), commonly referred to as the Psyche Mission Dictionary, was designed during the projects phase C and D development, and the first cruise flight segment. The dictionary is based on the 1.22.0 PDS information model. Attributes can be found in the Discipline_Area section of a PDS4 label.

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CHAPTER

THREE

ORGANIZATION OF CLASSES AND ATTRIBUTES

The following classes and attributes are defined for the Psyche mission. SLCK values can be provide with or without a clock partition, and in either byte or decimal based notation. If the SCLK is given without a clock partition, the spacecraft_clock_partition keyword must be defined. Example of properly formatted SLCK values include: 0/0792731143-0277070, 0792731143-0277070, or 0792731143.2642345.

3.1 Mission Wide Classes and Attributes

The following attributes are defined for the mission (Psyche_Mission super-class), and found within the Psyche_Parameters class.

psyche:mission_phase_name - This class contains an enumerated text string corresponding to the specific mission phase that a product was acquired in. If a product spans a phase boundary, this attribute should be set to phase in which the observation began acquisition. *psyche:orbit* - This class contains an integer value corresponding to the orbit number, in that particular orbital operations phase. *psyche:release_number* - This class contains an integer value corresponding to the release number that a particular product was associated with. *psyche:spacecraft_clock_partition* - This class contains the integer value corresponding to the current mission clock partition for the current observation. *psyche:spacecraft_clock_start* - This class contains the SCLK value at the start of the observation. *psyche:spacecraft_clock_stop* - This class contains the SCLK value at the end of the observation. *psyche:spacecraft_clock_start_corrected* - This class contains the corrected SCLK value at the start of the observation. *psyche:spacecraft_clock_stop_corrected* - This class contains the corrected SCLK value at the end of the observation. *psyche:spacecraft_fsw_dictionary_version* - This class contains an enumerated string value corresponding to the flight software dictionary being used the observation acquisition. Acceptable values are: `'5_4_2_1' or '5_4_2_2'. [psyche:spacecraft_active_flight_computer](#spacecraft_active_flight_computer) - This class contains an enumerated character value corresponding to which RCE was in control during the observation acquisition. Acceptable values are:` A' or `B'.

3.2 Instrument Specific Classes and Attributes

As of version 1.1.0.0 of the mission dictionary, there is a defined Magnetometer_Parameters and Imager_Parameters class, each containing instrument specific attributes. These classes are contained within the super-class Psyche_Mission.

3.2.1 Magnetometer Instrument Attributes

psyche:magnetometer_instrument_id - This attribute contains the enumerated magnetometer ID for the observation. *psyche:electronics_unit_id* - This attribute contains the mag electronics unit id for the observation. *psyche:magnetometer_temperature* - This attribute contains the mag temperature for the observation. *psyche:magnetometer_frequency* - This attribute contains the mag frequency for the observation. *psyche:magnetometer_delay* - This attribute contains the mag delay value for the observation. *psyche:magnetometer_sampling_frequency* - This attribute contains the mag sampling frequency for the observation.

3.2.2 Imager Instrument Attributes

psyche:imager_instrument_id - This attribute contains the enumerated imager ID for the observation.

CHAPTER

FOUR

<PSYCHE_MISSION_DICTIONARY>

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4.1 <psyche:>

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```
   <Discipline_Area>
     <psyche:Psyche_Mission>
       <psyche:Psyche_Parameters>
         <psyche:mission_phase_name>PHASE</psyche:mission_phase_name>
         <psyche:orbit>ORBIT_NUMBER</psyche:orbit>
         <psyche:release_number>RELEASE_NUMBER</psyche:release_number>
         <psyche:spacecraft_clock_partition>SCLK_PARTITION</psyche:spacecraft_clock_
↪partition>
         <psyche:spacecraft_clock_start>SCLK_START</psyche:spacecraft_clock_start>
         <psyche:spacecraft_clock_stop>SCLK_STOP</psyche:spacecraft_clock_stop>
         <psyche:spacecraft_clock_start_corrected>SCLK_START_CORRECTED</
↪psyche:spacecraft_clock_start_corrected>
         <psyche:spacecraft_clock_stop_corrected>SCLK_STOP_CORRECTED</psyche:spacecraft_
↪clock_stop_corrected>
         <psyche:spacecraft_fsw_dictionary_version>SC_FSW_DICTIONARY_VERSION</
↪psyche:spacecraft_fsw_dictionary_version>
         <psyche:spacecraft_active_flight_computer>SC_ACTIVE_COMPUTER</psyche:spacecraft_
↪active_flight_computer>
       </psyche:Psyche_Parameters>
       <psyche:Magnetometer_Parameters>
            <psyche.magnetometer_instrument_id>MAG_INST_ID</psyche.magnetometer_
↪instrument_id>
            <psyche.electronics_unit_id>MAG_EU_ID</psyche.electronics_unit_id>
            <psyche.magnetometer_temperature>MAG_TEMP</psyche.magnetometer_temperature>
            <psyche.magnetometer_frequency>MAG_FREQ</psyche.magnetometer_frequency>
            <psyche.magnetometer_delay>MAG_DELAY</psyche.magnetometer_delay>
            <psyche.magnetometer_sampling_frequency>MAG_SAMP_FREQ</psyche.magnetometer_
↪sampling_frequency>
       </psyche:Magnetometer_Parameters>
       <psyche:Imager_Parameters>
            <psyche.imager_instrument_id>IMG_INST_ID</psyche.imager_instrument_id>
       </psyche:Imager_Parameters>
```

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```
    <psyche:Psyche_Mission>
  </Discipline_Area>
```

4.1.1 Mission Phase

This parameter is enumerated to a known phase name, corresponding to specific date ranges of the mission.

- test The Psyche mission phase used for general testing purposes.
- preflight The Psyche mission phase used for preflight operations.
- atlo The Psyche mission phase used for ATLO.
- launch The Psyche mission phase used for spacecraft launch.
- ico The Psyche mission phase used for the initial checkout (ICO), lasting 100 days.
- cruise_1 The Psyche mission phase used for the cruise 1 period, lasting 26 months.
- mga The Psyche mission phase used for the Mars gravity assist, lasting 62 days.
- cruise_2 The Psyche mission phase used for the cruise 2 period, lasting 36 months.
- approach The Psyche mission phase used for the approach to 16 Psyche, last 100 days.
- orbital_operations bThe Psyche mission phase used for the entire orbital operations period, lasting 26 months.
- orbit_a The Psyche mission phase used for orbit A operations, lasting 56 days.
- orbit_b1 The Psyche mission phase used for orbit B1 operations, lasting 92 days.
- orbit_b2 The Psyche mission phase used for orbit B2 operations, lasting 100 days.
- orbit_c The Psyche mission phase used for orbit C operations, lasting 100 days.
- orbit_d The Psyche mission phase used for orbit D operations, lasting 100 days.
- orbit_transfer_a_to_b1 The Psyche mission phase used for orbit transfer A-to-B1, lasting 17 days.
- orbit_transfer_b1_to_d The Psyche mission phase used for orbit transfer B1-to-D, lasting 98 days.
- orbit_transfer_d_to_c The Psyche mission phase used for orbit transfer D-to-C, lasting 90 days.
- orbit_transfer_c_to_b2 The Psyche mission phase used for orbit transfer C-to-B2, lasting 23 days.

4.1.2 Flight Software Dictionary

This parameter is enumerated to known version of the flight software dictionary in use during the mission.

- 5_4_2_1 Version 5.4.2.1 of the Psyche FSW dictionary.
- 5_4_2_1 Version 5.4.2.2 of the Psyche FSW dictionary.
- 6_0_0_4 Version 6.0.0.4 of the Psyche FSW dictionary.
- 6_0_0_5 Version 6.0.0.5 of the Psyche FSW dictionary.
- 6_0_0_6 Version 6.0.0.6 of the Psyche FSW dictionary.
- 6_0_0_7 Version 6.0.0.7 of the Psyche FSW dictionary.
- 6_0_0_8 Version 6.0.0.8 of the Psyche FSW dictionary.
- 6_0_0_9 Version 6.0.0.9 of the Psyche FSW dictionary.

- 7_0_0_9 Version 7.0.0.9 of the Psyche FSW dictionary.

4.1.3 Magnetometer Instrument ID

This parameter is enumerated to the known versions of the flight magnetometer instrument suite.

- magnetometer_1 Psyche Magnetometer 1, NAIF ID -255170
- magnetometer_2 Psyche Magnetometer 2, NAIF ID -255180

4.1.4 Magnetometer Electronics Unit

This parameter is enumerated to the known versions of the flight magnetometer electronic units.

- EU_1 Psyche Magnetometer Electronics Unit 1
- EU_2 Psyche Magnetometer Electronics Unit 2

4.1.5 Imager Instrument ID

This parameter is enumerated to the known versions of the flight imager instrument suite.

- imager_a Psyche Multispectral Imager A, NAIF ID -255110
- imager_b Psyche Multispectral Imager B, NAIF ID -255130